

INDUSTRY DEMANDS

maintenance cost cited by NPCIL for setting up BSRs, levy of an exorbitant expertise fee at 60 paise/KWH of nuclear power generated, which translates into ₹150 crore per project, lack of financing guarantee with assets remaining under NPCIL's control, and challenges of land availability. They have also sought single window clearances and freedom to decide tariffs.

India's nuclear dreams now hinge on the government's response to these concerns. There have been a series of consultations at various levels in the government. The NITI Aayog has also been holding talks with the industry players who will help India achieve its 2047 power goals. The question is: How far is it willing to go?

IS IT VIABLE?

60 | According to documents accessed by *Business Today*, the industry players have sought a number of changes to the RFP, floated on December 31, 2024, saying conditions make nuclear projects unviable.

Though the government is engaging with the industry to ease some of these conditions and looking at the qualifying criteria for companies which will operate nuclear power plants, experts who track the sector say not all demands can be met due to the safety and security aspects of operating nuclear reactors.

In a change in stance, India is mulling to allow private players to mine, import, and process uranium, while reprocessing for fuel usage in nuclear power plants will remain with the government.

Vivek Jain, Director, India Ratings & Research, says a nuclear reactor cannot be a private asset in the true sense of the word. "In India, the model where you set up the asset, own it, and then transfer it, is un-



Lower operation and maintenance costs



Waive expertise fees @60 paise/kWh



Increase plant load factor to 90% and above



Allow foreign small modular reactor technology

likely to be adopted. The ownership will rest with a government agency or some other body, may be a joint venture. It (the asset) will never be under the direct control of private hands," he says. The companies say this will prevent them from raising funds from financial institutions. Only a few countries allow private players to own nuclear assets.

That is not all. The industry is looking at permission to set up latest reactors. It wants foreign technology—Gen III and Gen IV small

modular reactors. It has also requested that the civil nuclear liability and insurance cost be managed by NPCIL as it is well-equipped to handle the complexities of nuclear liability and insurance. The government is already looking at easing the Civil Liability for Nuclear Damage Act, 2010, which had a mechanism for compensating victims of a nuclear accident. The law was criticised by a number of foreign companies as an impediment to setting up nuclear power plants in the country.

India has 25 reactors with a capacity of 8.8 GW. Eighteen reactors, with a total capacity of 13.6 GW, including the 500 MW prototype fast breeder reactor, are at different stages of development. These will take the installed nuclear power capacity to 22 GW by 2032.

However, this is not enough. For the 2047 target, the government will have to get the private sector on board, which is turning out to be a bit of a challenge.

NUTS & BOLTS

NPCIL has received nearly 700 queries from dozens of companies, including Adani Energy Solutions, Aditya Birla Renewables, Paperboards Specialty Papers Division, ITC, Jindal Nuclear Power Pvt Ltd, JSW Energy, Reliance Industries and Tata Power Company. Adani Energy Solutions has said the operational expenditure cited by NPCIL is too high and will make the project unviable. "The operational expenditure of thermal power plants is ₹20-22 crore per MW, and the same has reduced over a period. It is also noted that the operational expenditure of nuclear power plants, in a report by the Danish Energy Agency and the Central Electricity Authority, is around ₹43 lakh/MW, likely to fall further.